P530/2
BIOLOGY
(Theory)
Paper 2
July/Aug.2023
2 1/2 hours

Uganda Advanced Certificate of Education

BIOLOGY

(THEORY)

Paper 2

2 hours: 30 minutes

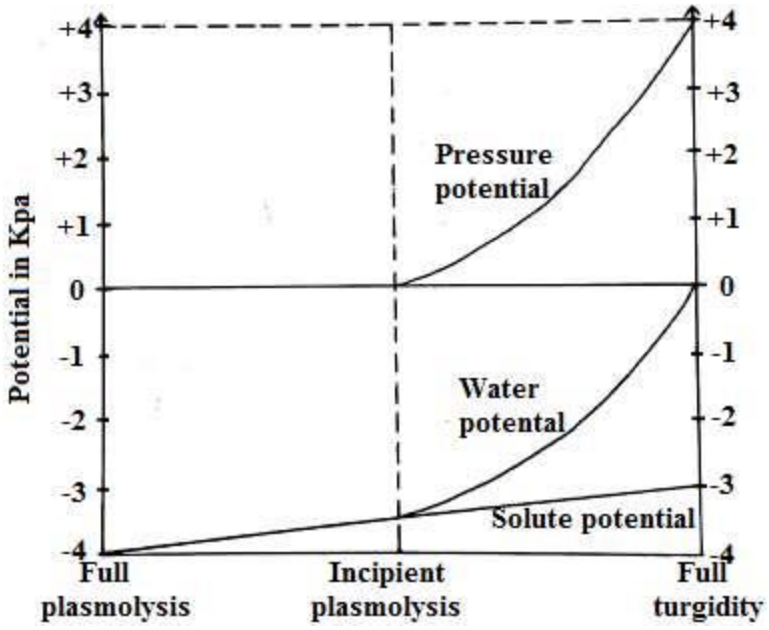
INSTRUCTIONS TO CANDIDATES:

- *This paper consists of **six** questions.*
- *Answer **one** question in section **A** plus **three** others from section **B**.*
- *Candidates are advised to read the questions carefully, organise their answers and present them precisely and logically, illustrating with well labelled diagrams where necessary.*

SECTION A (40 MARKS)

1. The graph below shows changes in water potential, pressure potential and solute potential of a full plasmolysed plant cell in a hypotonic solution.

Figure 1



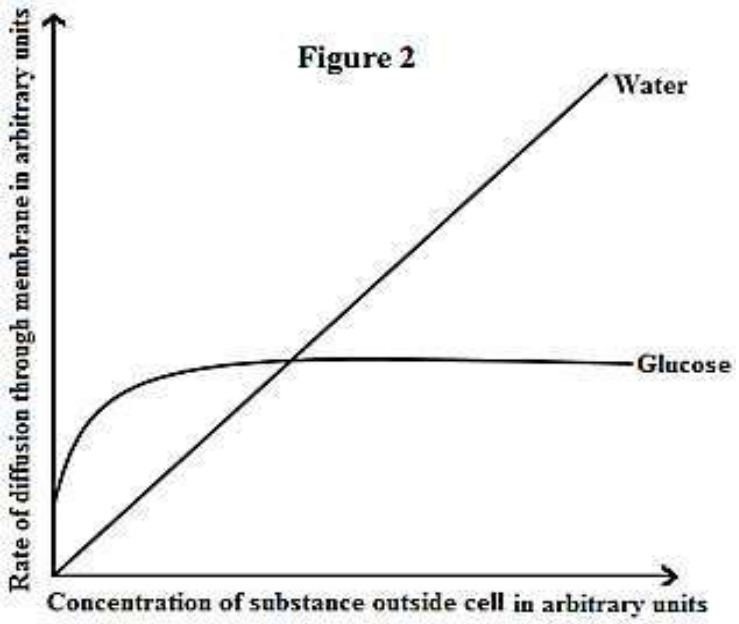
(a) Describe and explain the changes in;

(i). Water potential ***(09 marks)***

(ii). Solute potential ***(02 marks)***

(iii).Pressure potential ***(06 marks)***

Figure 2 shows the rate of movement of two different substances across a phospholipid membrane; glucose by facilitated diffusion and water by simple diffusion, at varying extracellular concentration.



(b). Describe the effect of increasing extracellular concentration on rate of:

(i) Glucose uptake. ***(06 marks)***

(ii) Water uptake ***(03 marks)***

(c).Explain the effect of increasing extracellular concentration on rate of:

(i) Glucose uptake ***(05 marks)***

(ii) Water uptake ***(04 marks)***

(d) With reason(s) suggest what would happen to the rate glucose uptake if the phospholipid bilayer is treated with potassium cyanide ***(05 marks)***

SECTION B (60 MARKS)

2. (a) Explain how the plasma membrane is modified to suit its function(s) ***(10 marks)***

   (b) Describe the role of intracellular membranes to an eukaryotic cell. ***(10 marks)***

3. (a) Describe factors for existence of a variety of carbohydrates in nature. ***(10 marks)***

   (b) Explain the biological significance of physical properties of water to organisms ***(10 marks)***

4. (a) Describe how the structure of tracheids relate to its function ***(06 marks)***

   (b) How is the xylem vessel formed from its meristematic cells ***(09 marks)***

(c) Give five characteristics of meristematic tissues in higher plants ***(05 marks)***

5. (a) Describe changes which occurs in the nucleus during meiosis ***(12 marks)***

(b) Explain the importance of the following to organisms

(i) Meiosis ***(04 marks)***

(ii) Mitosis ***(04 marks)***

6. (a) Describe the structure of the stoma ***(04 marks)***

(b) Explain how osmotic pressure changes control stomatal movements ***(10 marks)***

(c) Describe how water ascends the plant with a cut shoot from soil solution ***(06 marks)***

END ©MS